Quiz 5: Liquids & pH

Review of Session 5. ~10–15 minutes.

Section A: pH

1. What does pH stand for?
2. If a liquid has a low pH, what does it have a higher concentration of?
3. The pH scale is logarithmic. How many times more acidic is pH 3 than pH 5?
4. What pH is neutral? What range is acidic? What range is basic?
5. Acid + base → ______ + ______.

Section B: The liquids

Give the formula and approximate pH:

1. Water
2. Vinegar
3. Household ammonia
4. Hydrogen peroxide
5. Isopropyl (rubbing) alcohol

Section C: Identification

1. Two clear liquids are both sour-smelling and read pH 2.5. One is cloudy and yellowish. Which is which?
2. A mystery liquid foams when a drop lands on a piece of raw potato or blood. What is it, and what enzyme causes the foam?
3. A liquid evaporates quickly off the back of your hand and smells medicinal. What is it?
4. Two neutral, odorless liquids. One dries to cubic crystals, the other dries sticky with no crystals. Name each.
5. Red cabbage indicator turns bright pink in a liquid. Acid or base? Roughly what pH?

Answer Key

Section A

1. "Potential of hydrogen" (or "power of hydrogen").
2. Hydrogen ions (H^+).
3. 100 times (2 units × 10 each).
4. Neutral = 7; acidic = below 7; basic = above 7.
5. salt + water.

Section B

1. H_2O, pH 7.
2. CH_3COOH (or $C_2H_4O_2$), pH 2–3.
3. NH_3, pH 11–12.
4. H_2O_2, pH ~4–6.
5. C_3H_8O, pH ~6–7.

Section C

1. Cloudy and yellowish = lemon juice; clear = vinegar.
2. Hydrogen peroxide; catalase.
3. Isopropyl alcohol.
4. Cubic crystals = salt water (NaCl); sticky, no crystals = sugar water.
5. Acid; about pH 3–4.